Large Language Models in Human-machine interaction

Ömer Emre Mutlu

emre.mutlu@rwth-aachen.de

Abstract

This document serves as a template for the seminar write-up on Large Language Models in human-machine interaction. It will be replaced with the final abstract once the paper is complete.

Keywords: Large Language Models, Human-Machine Interaction, Seminar

CONTENTS

Disclaimer: Artificial intelligence tools were used strictly for grammar correction and minor editorial polishing; all analysis, arguments, and original content remain solely the work of the author.

1 Introduction

In high-stakes environments such as aerospace, medicine, disaster response, military applications, and similar domains, the effectiveness and speed with which operators interact with the tools available to them are of critical importance. Traditionally, human–machine interaction (HMI) in such domains has been achieved through methods that have been field-tested over many years, such as user interfaces (UIs) utilizing buttons, joysticks, haptic-feedback sensors, and touchscreens [RSS+25]. In regulated application domains specific norms and regulations govern the permissible interaction mediums for HMI user interfaces, including standards such as IEC 60204-1 for industrial electrical equipment and NASA-STD-3001 for aeronautical systems.[Int16][CWF23]

However, with the introduction of transformer encoder–decoder models and the subsequent popularization of large language models such as Google's BERT, OpenAI's GPT-3, and more recently GPT-5, a wide range of new possibilities has emerged in the field [CYJ+25]. Moreover, recent advancements in large language models in the areas of speech-to-speech and visual language models have created additional and particularly interesting opportunities [CFD24].

This paper presents a literature review of recent advancements in relation to human–machine interaction within the context of safety-critical working environments.

2 Background

Before proceeding with the analysis of domain-specific work, a brief introduction to the involved technologies is necessary to provide the reader with sufficient background and contextual understanding. Accordingly, this section presents a concise overview of the relevant technologies.

In a human–machine interaction (HMI) system, the human can be regarded as a component equipped with four primary input–output channels: vision, hearing, touch, and motor activity (movement) [DFAB04]. Among these channels, vision and hearing are most commonly addressed in contemporary AI-driven interaction systems and are therefore particularly relevant in the context of large language model (LLM) based interfaces. For the visual modality, current approaches primarily involve text-based language models and vision–language models that process textual input alongside visual information. For the auditory modality, interaction is typically realized through audio-based processing pipelines that combine speech recognition and speech synthesis with language models. The subsequent subsections further examine these modalities in more detail.

2.1 Text-Based LLM Interfaces

Since the introduction of the Transformer architecture in "Attention Is All You Need", text-based content generation and processing have become the central foundation of modern natural language processing (NLP).[VSP+17] In this context, large language models are a family of neural language models trained on vast amounts of textual data and capable of performing a wide range of natural language processing tasks, such as generating context-aware responses based on user-provided text prompts.[BMR+20] To utilize this family of models as a user interface, the model must be capable of understanding domain-specific context and of acting upon user prompts, rather than merely generating text responses. Several methods exist to enable this, a brief summary of these approaches is provided below.

Fine Tuning

Fine-tuning is the process of further training a pre-trained model on task- or domain-specific data by updating its existing parameters or, in some approaches, by introducing additional trainable parameters.[Mos23] By further training the model on a domain-specific dataset, for example technical data on extraterrestrial habitats, a base model such as GPT-5 can acquire contextual knowledge of the application domain and generate its responses accordingly.

However, a significant downside of this approach is the increased risk of model hallucinations. Hallucinations refer to the phenomenon in which a model generates incorrect or fabricated information in response to a given prompt.[GYA+24] Such behavior can pose a significant risk in safety-critical environments.

Retrieval Augmented Generation

Retrieval-Augmented Generation (RAG) introduces relevant external information to the language model without modifying the model's internal parameters.

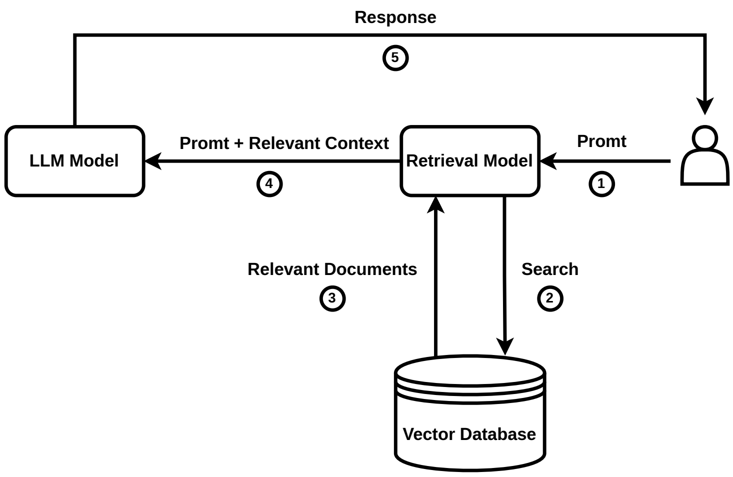


Figure 1: High-level RAG workflow that injects retrieved documents into the prompt instead of altering model weights.

As shown in Figure 1, the RAG workflow begins with a user prompt; however, instead of passing the prompt directly to the LLM, a similarity search is first performed over context-relevant information stored as embeddings in a vector database. The retrieved context is then provided together with the prompt to the LLM for grounded response generation.[BMR+20] Recent studies have shown this to be a more reliable method for context retrieval compared to fine-tuning.[OBME24]

However, the risk of hallucinations remains non-zero even with RAG; nevertheless, with well-designed context selection and retrieval strategies, this probability is significantly reduced, as demonstrated in numerous studies [JLF$^+$24].

Tool Use / Function Calling

Tool calling, also referred to as function calling, is the ability of a large language model to invoke external functions by interpreting user context and generating structured Application Programming Interface (API) calls with the relevant parameters extracted from user prompts, as introduced by Schick et al. [SDYD$^+$23].

In a practical setting, this means that the model generates structured tool calls, which are then used by a service layer to invoke HTTP requests to downstream services for task execution, as illustrated in Figure 2. By utilizing this method, the model can interact with external systems to trigger the execution of real-world functions.

In the context of human–machine interaction, this implies that a user can issue natural-language prompts to control robots and machines to execute a wide variety of tasks, as demonstrated by Vemprala et al. [VBBK23].

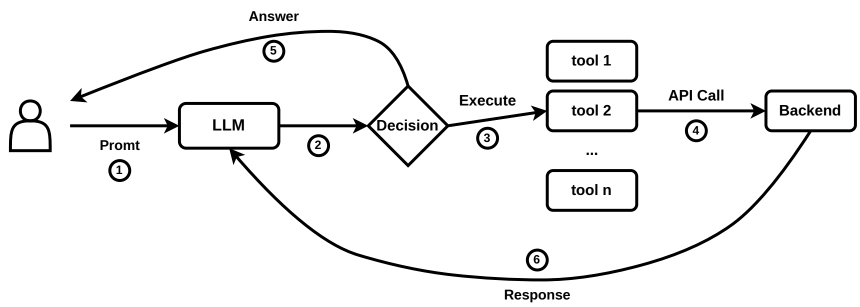


Figure 2: Conceptual workflow of tool (function) calling in a large language model.

Agentic Artificial Intelligence

What is commonly referred to as agentic AI is the combination of previously discussed methods to enable semi-autonomous decision-making systems. An AI agent typically integrates a large language model for reasoning and planning with tool-calling capabilities and memory in order to execute tasks without constant human supervision, requesting user input only when necessary to adjust its actions [YZY$^+$23].

Such agents are currently most commonly used in programming-related tasks; a prominent example of an agentic AI system in this domain is OpenAI's Codex-based coding agent.

With regard to the use of such systems in human–machine interaction, Kim et al. note that the application of agentic systems in this context remains underexplored. However, their user study reveals that current agentic systems are still prone to failure, which caused usability issues for some participants [KLM24]. The agents in their experiment exhibited persistent hallucinations, leading some participants to doubt their own knowledge. Additionally, in cases where a physical robot embodiment was used to represent the agent, participants reported an uncanny-valley-like sense of unease during task execution. Nevertheless, the field and the underlying technology are still at an early stage of development, and user experience is expected to improve over time. The application of such systems in domains such as deep-space exploration remains a particularly promising research direction, as discussed by Heinicke et al. [FSG+21]. However, in their current state, the suitability of such systems for safety-critical applications remains uncertain.

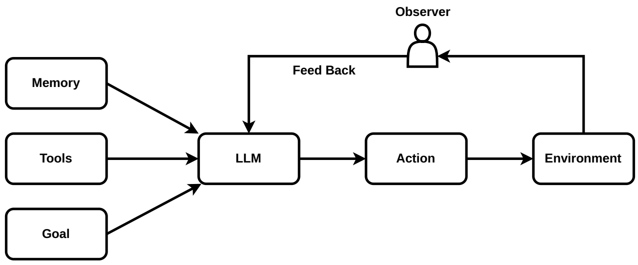


Figure 3: High-level agentic AI stack combining planning, memory, tool use, and execution loops.

2.2 Audio Based Interfaces

A critical component of human–machine interaction is speech, which represents one of the most natural and powerful means of human communication. In many HMI application domains, the ability to translate spoken user input into corresponding machine actions is of particular importance. Accordingly, this section discusses two methods that are currently employed to enable speech-driven interaction in such systems.

Speech-to-Text, LLM and Text-to-Speech

As highlighted by Haeb-Umbach et al. and other previous work, until recently this architecture represented the primary approach for providing voice support in interactive human–machine interaction (HMI) applications [HUWN+]. The overall pipeline typically consists of a speech detection and preprocessing layer, which forwards the audio signal to a speech-to-text (STT) model. The spoken input is thereby converted into text and subsequently processed by a large language model (LLM), which performs the reasoning and retrieval operations introduced in Section 2, such as retrieval-augmented generation (RAG). The LLM then produces a textual response, which is finally converted back into speech and presented to the user via a text-to-speech (TTS) system.

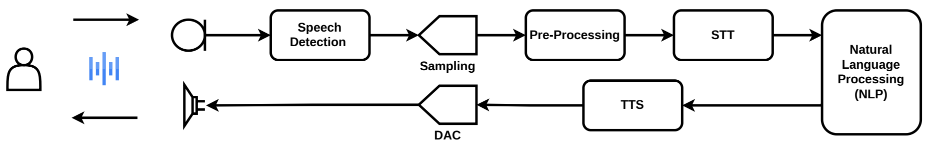


Figure 4: Signal chain for speech-to-text ingestion, LLM reasoning, and text-to-speech playback.

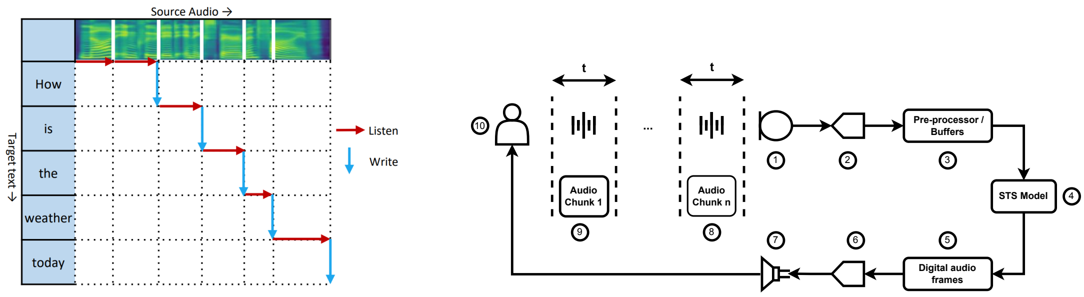
However, this pipeline also exhibits notable drawbacks. The sequential use of speech-to-text (STT), large language model (LLM) processing, and text-to-speech (TTS) synthesis can introduce perceptible latency. To mitigate this effect, some systems employ interaction design strategies such as playing short default utterances (e.g., “Let me think about that…”) while the main response is being generated. Nevertheless, empirical user studies indicate that such waiting times can still negatively affect the naturalness and fluidity of human–machine interaction, particularly in conversational settings [FSG+21, KLM24].

On the positive side, such a configuration allows spoken input to be converted into textual form, which can facilitate integration with retrieval-augmented generation (RAG) systems and thereby support more efficient retrieval of contextually relevant information.

Speech-to-Speech LLMs

Direct end-to-end speech-to-speech models represent a relatively recent development within the broader family of large language and multimodal models. Unlike the cascading approach discussed in the previous section, these models do not rely on an explicit intermediate text representation; instead, linguistic processing is performed within a model trained directly on speech signals [SSJ+25]. This design can reduce end-to-end latency and mitigate error propagation that may arise from multiple processing stages in traditional multi-stage pipelines, and can support more natural conversational interaction with the system .

Within end-to-end speech-to-speech systems, two broad subcategories can be distinguished: direct models and real-time (streaming) models. Direct models process speech input as complete utterances, generating output only after the full input signal has been received. Their interaction pattern resembles cascading pipelines in terms of latency, as response generation is deferred until the entire input has been processed [ZFG+24].



(a) Streaming wait-k policy alternating between listening to additional source frames and emitting target tokens in real time [TLR+20].

(b) Streaming signal chain diagram with numbered legend.

Legend: 1) Microphone interface capturing the operator's raw speech
2) Low-latency Analog to Digital Converter
3) Pre-processing buffers that normalize, filter, and batch frames
4) Streaming STS model performing NLP task
5) Generated digital audio frames for the target utterance
6) Vocoder/decoder producing analog waveforms
7) Speaker/headset driver for immediate playback
8 - 9) The Audio is processed in chunks that are (t) in length, usually in the order of 20-40ms
10) End user simultaneously providing input and receiving translated speech.

Figure 5: Real-time speech-to-speech pipeline: (a) wait-k listen/write policy and (b) corresponding signal path.

In contrast, more recent real-time speech-to-speech systems, such as OpenAI's real-time speech model family adopt a streaming-based approach. Incoming speech is first digitized (e.g., using pulse-code modulation formats such as PCM16) and segmented into short audio frames on the order of tens of milliseconds. These segments are streamed incrementally to the model, which buffers a limited temporal context before beginning to generate spoken output [Ope24].

In Figure 5a, an example of such a system is illustrated using the simultaneous speech model introduced by Tan et al., showing how these models interleave listen and write operations according to a wait-k policy, allowing target-language output to begin before the full source utterance is observed [TLR$^+$20]. To ground this abstract description in the physical signal chain used in safety-critical HMI deployments, Figure 5b annotates a representative streaming speech-to-speech path from the operator's microphone through synthesis and playback.

This streaming formulation enables incremental processing and response generation, which can reduce perceived latency and support a more natural conversational flow compared to full utterance-level processing approaches. However, a downside of such an approach is that end-to-end speech-to-speech models typically require larger computational resources compared with cascaded systems discussed previously, and they can face challenges in generalizing across diverse domains due to model architecture. Additionally, as of late 2025, native retrieval-augmented generation (RAG) compatibility remains an active area of research for end-to-end speech models, with most practical RAG implementations still relying on intermediate text representations for effective document retrieval [MML$^+$25].

2.3 Vision-Language Models

Vision is another input channel that is of central importance to human psychological perception and interpretation, particularly in tasks where spatial reasoning and situational awareness are critical. Vision–language models extend machine interaction capabilities by enabling the joint processing of visual inputs and natural language, thereby allowing contextual understanding and reasoning grounded in visual information. These models exist at the intersection between computer vision and natural language processing [LWD$^+$25]. For example, such models can be applied to the analysis of human facial expressions to infer affective states from visual data. An illustrative example of this type of application is provided in an accompanying notebook.[1]

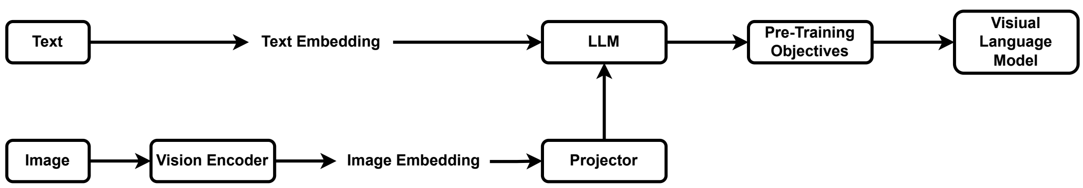


Figure 6: Conceptual VLM workflow: the vision encoder produces embeddings that are aligned and fused with LLM tokens to support multimodal reasoning and grounded responses.[LWD$^+$25]

Vision-Language Models (VLMs) work by using a vision encoder to convert images into numerical feature representations and a large language model (LLM) to convert text into contextual embeddings. The vision encoder extracts visual features, which are then transformed into a compatible format before being integrated with the language model's input. In many modern

[1] `https://github.com/EmreMutlu99/Large-Language-Models-in-Human-machine-interaction/blob/main/tests/visual-language-models.ipynb`

VLMs, a pre-trained LLM serves as the backbone, and a projection or fusion layer is used to inject visual embeddings into the LLM so it can process visual information alongside text. This enables multimodal understanding and generation. A representation of this process is provided in Figure 6 [LWD+25]. In the context of human–machine interaction (HMI), such models can be deployed alongside audio- and text-based models to enhance the overall user experience in interactive systems.

3 Domain Case Studies

In the previous section, the architectural foundations underlying the use of large language models (LLMs) in the context of human–machine interaction were introduced, encompassing text-based, audio-based, and vision–language modalities. This section builds on that foundation by reviewing selected studies from several safety-critical application domains; namely aerospace and remote sciences, emergency and environmental response, and industrial operations, with a focus on how LLM-based technologies are applied in human–machine interaction within these contexts.

3.1 Aerospace and Remote Science

Aerospace and remote science missions place humans in some of the most demanding HMI settings. Astronauts and mission operators work under high risk, limited communication, and long periods of isolation, while managing complex systems and strict safety procedures. In these conditions, interaction approaches that reduce cognitive effort, support decision-making, and offer context-aware assistance are especially valuable. In this section, selected papers from this domain are analyzed and discussed.

Conversational User Interfaces to Support Astronauts [FSG+21]

This paper uses a Wizard of Oz (WoZ) study[2] to investigate how astronauts would realistically interact with a conversational user interface (CUI) inside an extraterrestrial habitat before such a system technically exists. It investigates how CUIs can support astronauts living and working in extraterrestrial habitats (Moon/Mars), where isolation, confinement, delayed communication with Earth, and psychological stress are major challenges.

To address these differing interaction requirements, the authors designed two complementary CUIs: CASSIOPEIA, which provides task-oriented, mission-critical assistance for scientific operations, and PEGASUS, which functions as an emotionally expressive personal companion to support crew well-being.

While designing the conversational user interfaces (CUIs), the authors identified four key design guidelines. First, CUIs should support both mission-critical and non-mission-critical interactions, ensuring reliable task execution while also accommodating informal and supportive exchanges. Second, the interfaces should be designed for frequent and prolonged use, taking into account long-duration missions and sustained human–AI interaction without causing cognitive fatigue. Third, CUIs should enable multimodal and bidirectional interaction, allowing users to communicate with the system through complementary input and output modalities such as text,

[2] A Wizard of Oz study is an experimental method in which users interact with an apparently autonomous system whose behavior is secretly controlled by a human operator, which enables the study of interaction patterns before full system implementation.[DJA93]

speech, and visual feedback. Finally, the CUI should be designed as a team member, positioned as an active collaborator that supports coordination, shared situational awareness, and trust within the crew.

Method

To derive requirements for conversational user interfaces in extraterrestrial habitats, the authors conducted a week-long Wizard-of-Oz study inside the MaMBA Moon/Mars habitat laboratory module at ZARM. Four experienced scientists carried out real biological, geological, and materials science experiments while interacting with a simulated CUI operated by a human wizard via a Raspberry Pi–based voice interface. All interactions were logged, video-recorded, transcribed, and qualitatively coded, revealing frequent use of both mission-critical and non-mission-related interactions. Based on these findings, the authors derived four design guidelines emphasizing support for mission and non-mission tasks, intensive long-term use, multimodal and bidirectional interaction, and the treatment of the CUI as a team member. Guided by these principles, the two CUIs introduced earlier were implemented. These systems were evaluated in two controlled user studies within the habitat: one comparing voice-only versus voice-plus-GUI interaction for task efficiency, and another comparing emotionally characterized versus neutral CUIs to assess emotional engagement, stress, and workload using standardized questionnaires and interviews.

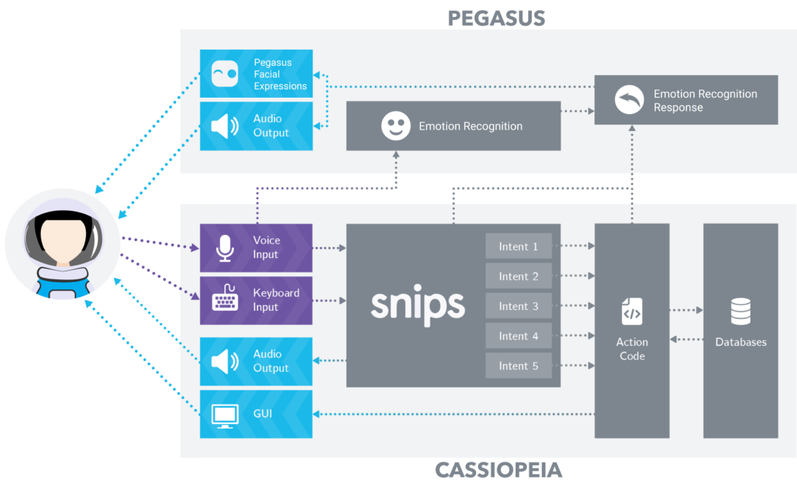


Figure 7: Overview of the PEGASUS and CASSIOPEI companion CUIs: multimodal astronaut inputs are parsed through intent recognition, emotional context, and action code to provide both operational and affective support.[FSG+21]

Result

The study shows that CUI can meaningfully support astronauts in isolated extraterrestrial habitats by improving both task performance and user experience. Results indicate that multimodal CUIs combining voice and graphical interfaces enable significantly more efficient task completion than voice-only CUIs, while perceived usability remains comparable. Emotionally characterized CUIs increase user excitement and engagement, and stronger emotional connection and perceived sympathy toward the system are associated with lower reported stress levels, although individual responses vary and overall task workload does not differ significantly. Overall, the findings suggest that well-designed CUIs can enhance operational efficiency and psychological well-being in long-duration space missions, provided that emotional interaction and interface modality are carefully adapted to user needs.

Analysis

Freitas et.al. propose a framework for the design and development of conversational user interfaces for space missions in this paper. Although the study employs a Wizard-of-Oz setup in which a human operator simulates the AI system, the results provide a convincing proof of concept demonstrating that the development of such assistants is both feasible and worthwhile. Building on the proposed framework, a fully autonomous implementation that integrates real-time speech-to-speech models, retrieval-augmented generation, and tool-calling mechanisms represents a promising direction for future research. Such an approach could combine several aspects discussed in the background section of this paper to realize an agentic conversational system capable of supporting astronauts in safety-critical and isolated mission environments.

AI Assistants for Spaceflight Procedures: Combining Generative Pre-Trained Transformer and Retrieval-Augmented Generation on Knowledge Graphs With Augmented Reality Cues [BBN$^+$24]

In this paper Bensch et.al. intoduce an intelligent personal assistant (IPA) they named CORE (Checklist Organizer for Research and Exploration). CORE is designed to support astronauts during procedural tasks on the ISS, Lunar Gateway, and future deep-space missions. It combines Large Language Models (GPTs) with Retrieval-Augmented Generation (RAG), Knowledge Graphs (KGs), and Augmented Reality (AR) to achieve its tasks.

Method

The authors validate CORE through qualitative, prototype-based functional testing rather than formal benchmarking. Ten publicly available ISS procedures were indexed into a vector database and linked with metadata and images via a knowledge graph, forming the basis of a retrieval-augmented generation (RAG) pipeline built on GPT-4. Spoken user queries were transcribed using Whisper, enriched with retrieved procedure content, and used to generate constrained responses. Validation was performed using example queries to verify that CORE returned the correct procedure steps verbatim and triggered associated visual cues, while rejecting non-procedure-related questions. In addition to this functional validation, offline availability is addressed as a system design consideration: the authors argue that recent advances in open-source large language models, together with optimization techniques such as low-rank adaptation and quantization, make fully offline deployment of the proposed architecture feasible for future missions. However, this offline capability is not empirically evaluated in the present work. No quantitative performance metrics or user studies are reported; a controlled evaluation at the European Astronaut Centre is explicitly deferred to future work.

Result

The authors demonstrate that the CORE prototype can correctly retrieve and present procedural information using a Retrieval-Augmented Generation architecture. In their experiments, CORE successfully returned exact, verbatim procedure steps in response to user queries, including the correct step numbering and associated visual references when available. The system was able to answer both step-specific questions (e.g., requesting a particular step in a CPR procedure) and metadata-related queries (such as the last update date of a procedure), reformulating retrieved graph data into coherent natural-language responses. Importantly, CORE consistently rejected questions unrelated to the indexed procedures, enforcing strict domain boundaries and reducing the risk of hallucinated or unsafe responses. These results indicate that the proposed combination of GPT, RAG, and knowledge graphs can support reliable, constrained procedural assistance. However, the reported results are illustrative and qualitative, and no quantitative performance metrics, comparative baselines, or user-study outcomes are provided.

Analysis

Bensch et al. demonstrate that the development of an AI-based procedural assistant for space missions is technically feasible. They adopt a modular speech-to-text, language model, and text-to-speech (STT–LLM–TTS) pipeline, which enables seamless integration with a vector database through Retrieval-Augmented Generation (RAG). According to the authors, this architectural choice allows the system to retrieve reliable, procedure-specific information while reducing the risk of hallucinated responses. However, the paper lacks an in-depth evaluation of the system's capabilities; the limited number of illustrative prompt–response examples is insufficient to establish the robustness or generalizability of the approach. Furthermore, no user study is conducted to assess usability, user acceptance, or cognitive workload during interaction. As the system relies on a pipeline-based interaction model rather than an end-to-end speech-to-speech approach, response latency may be higher, potentially reducing conversational naturalness and introducing an "uncanny valley" effect. Despite these limitations, the authors' emphasis on offline-capable deployment represents a particularly important contribution, as reliance on cloud-based models is impractical in space environments due to communication delays and outages. This aspect highlights promising directions for future research, including the development of onboard data processing infrastructure capable of locally hosting large language models for autonomous space operations.

LLM-Based Autonomous Agent for Aerospace Wire Harnessing [WGZ+25]

In this paper, Wang et al. introduce an agent-based framework for human–robot collaboration (HRC) and validate it through a high-precision aerospace wire harnessing assembly use case. Building on existing tool-calling and agentic reasoning paradigms discussed in Section 2, the framework extends these approaches by incorporating long-term memory mechanisms, implemented via vector database embeddings, to support contextual understanding, and iterative task execution.

The proposed system is an agent-based human–robot collaboration (HRC) framework in which a large language model (LLM) is embedded into a robotic system as an autonomous decision-making agent. Rather than treating the LLM as a passive instruction parser, the framework elevates it to an agentic role, capable of reasoning, planning, executing actions through tools, observing outcomes, and adapting behavior over time. Human operators interact with the system primarily through natural language instructions, provided in text or speech form, while the robot executes physical actions and provides implicit feedback through its movements and task outcomes; continuous collaboration is achieved by allowing the human to issue incremental or corrective instructions during execution, which are interpreted by the agent using contextual memory and integrated into subsequent planning steps.

The architecture consists of several interconnected modules that together enable task-oriented autonomous behavior. The configuration module defines the agent's role, goals, behavioral constraints, available tools, and prompt structure, ensuring that the LLM operates as a task-oriented autonomous agent rather than a conversational chatbot. Task planning is handled by a dedicated module that uses an unmodified LLM within a Reason–Act chain-of-thought loop to interpret human instructions, decompose complex tasks, and determine which actions and tools should be invoked at each step. The task execution module then carries out the planned actions by invoking encapsulated tools for perception, coordinate transformation, and robot control, and feeds the execution outcomes back to the agent as observations. Finally, a memory module maintains both short-term contextual memory and long-term vector-embedded summaries of past interactions, enabling continuous and context-aware reasoning across multiple tasks

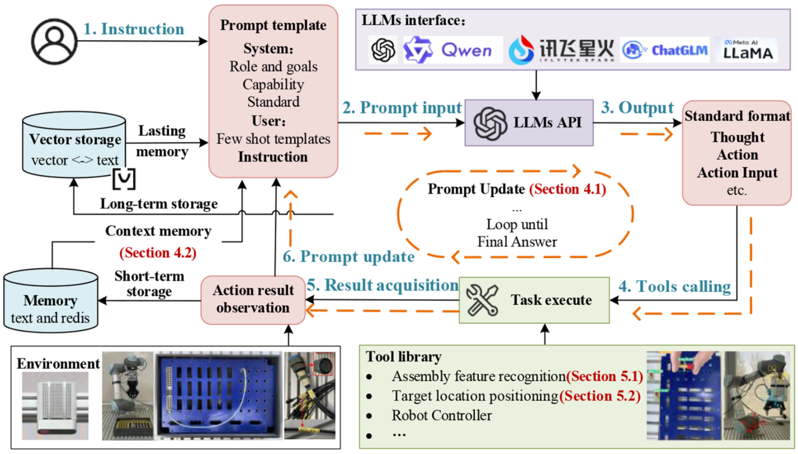


Figure 8: LLM-driven human–robot collaboration workflow showing how instructions propagate through prompting, memory, tool calls, execution, and feedback loops to support industrial assembly tasks. [WGZ+25]

Method

The proposed HRC agent is validated experimentally using a real-world human–robot collaborative assembly setup based on a UR5 collaborative robot equipped with a depth camera and vision system. Validation is conducted progressively across three capability levels: perception, task planning, and task execution. First, the agent's sensing ability is evaluated using a Target Recognition Success Rate (TRR), which measures the correctness of object detection and spatial localization for tools, connectors, cables, and assembly ports under repeated trials, including dense connector arrays handled via a grid-matching recovery method. Second, the cognitive task-planning capability is assessed independently of robot execution by providing natural-language instructions and evaluating whether the agent's chain-of-thought reasoning produces correct and executable action sequences, quantified using the Task Planning Success Rate (TPR) and benchmarked against state-of-the-art standalone LLMs under identical tool constraints. Third, full end-to-end task execution is validated by deploying the planned action sequences on the physical robot and measuring the Task Execution Success Rate (TER), requiring all subtasks to be successfully completed in real assembly scenarios. Each task is repeated multiple times to assess robustness, and additional experiments demonstrate the agent's generalizability to previously unseen wire assembly tasks, thereby validating perception, reasoning, and execution capabilities in an integrated industrial setting

Result

The proposed HRC agent demonstrates consistently high success rates across perception, task planning, and execution in real aerospace wire harnessing assembly experiments, with most tasks achieving 9–10 successful trials out of 10 repetitions. In task planning, the agent substantially outperforms standalone state-of-the-art LLMs, which frequently fail to generate correct and executable action sequences for domain-specific assembly tasks. Overall, the results indicate that the agent-based LLM framework enables robust and reliable human–robot collaboration in complex industrial environments

Analysis

The paper presents a technically solid and credible system, with results that are convincing within the tested industrial context, particularly because validation is performed on a real robotic setup rather than in simulation. Reported success rates (mostly 9–10 out of 10 trials) indicate robust performance, and comparisons with standalone LLMs support the claim that the agent-based architecture is the key contributor. However, the evaluation is limited in scope, lacks statistical analysis and ablation studies, and does not include human-centered metrics, so the results should be viewed as a strong proof of concept rather than evidence of broad generalizability.

Moreover, the system does not employ retrieval-augmented generation and instead relies on the language model's internal memory for context understanding, which limits access to explicit domain knowledge. Integrating a RAG mechanism to retrieve subject-specific assembly or process information would be a natural improvement to enhance robustness and scalability.

3.2 Emergency and Environmental Response

Large Language Models (LLMs) offer new opportunities to support emergency and environmental response by enhancing situational awareness and decision-making under time pressure. In this section, a general review of their use in this field is presented.

Large Language Model Applications in Disaster Management: An Interdisciplinary Review [XMLC25]

Xu et al. provide a systematic interdisciplinary review of how large language models (LLMs) are being applied across the disaster management domain, taking into account both technical foundations and practical implementations. The paper analyzes 70 research studies published between 2020 and 2024 to chart the evolving role of LLMs in disaster risk reduction, emergency response, and resilience-building efforts.

The review characterizes large language models as human centered intermediaries that are capable of processing vast amounts of heterogeneous, unstructured, and time critical information. This includes data sources such as social media content, emergency situation reports, sensor streams, and remote sensing observations. Within disaster management settings, these capabilities enable LLMs to support human decision makers operating under uncertainty, high time pressure, and rapidly evolving conditions. The authors identify three recurring limitations in current disaster management systems. First, existing solutions tend to focus predominantly on the response phase, while offering comparatively limited support for preparedness, mitigation, and recovery activities. Second, there is insufficient integration among diverse stakeholders and heterogeneous data sources, which restricts the formation of a shared and coherent operational picture. Third, although situational awareness is often achieved, it is frequently not translated effectively into concrete and actionable intelligence that can directly inform decisions. To address these challenges, the paper introduces the *3M Framework* as a conceptual foundation for next generation LLM driven disaster information management systems.

Method

The authors carried out a systematic literature review guided by the PRISMA[3] methodology, drawing on major academic and preprint databases including Web of Science, IEEE Xplore, ScienceDirect, Scopus, and arXiv. Following a two stage screening process, a total of 70 studies were selected for inclusion. These studies employ large language models or contextual language models such as GPT, BERT, and T5 within disaster related applications. The selected works were examined along two primary dimensions. The first dimension concerns the operational phase of disaster management addressed by each study, including detection, tracking, analysis, and action. The second dimension focuses on the technical approach, covering aspects such as the underlying model architecture, the chosen adaptation strategy, whether fine tuning or prompt engineering, and the degree of integration with external tools or additional data sources. This structured synthesis allows for a comparative perspective on how LLM capabilities are applied and evolve across different disaster types and across the various phases of disaster management.

Result

Large language models show strong performance in detecting disaster-related content, including misinformation filtering, humanitarian needs classification, and severity estimation from informal and multilingual data. Multimodal models further improve early warning by jointly processing text, images, and sensor data. For tracking, LLMs enhance event monitoring through entity extraction, temporal reasoning, and spatial interpretation; zero-shot methods reduce reliance on annotated datasets, though fine-grained disambiguation across locations and entities remains challenging. In analysis, LLMs support impact assessment via sentiment modeling, topic evolution, semantic networks, and knowledge graphs, enabling evaluation of response effectiveness and stakeholder communication. In the action phase, LLMs operate as interactive planning assistants, coordinating responders and affected populations through retrieval-augmented generation, tool orchestration, and scenario analysis.

Decoder-focused models, such as the GPT-4 family, increasingly dominate due to strong reasoning and generative capabilities, while encoder-based models remain important for low-latency and resource-constrained settings.

To guide future research and deployment, the authors propose the 3M Framework: multi-modal data fusion, integrating textual, visual, sensor, and structured data into unified situational assessments; multi-source information validation, cross-checking heterogeneous and conflicting sources to improve reliability under uncertainty; and multi-agent knowledge integration, coordinating human experts, AI agents, and physical–virtual systems to produce actionable intelligence.

Analysis

The authors present LLMs as agentic assistive tools that support operators in tasks such as planning, identification, and classification, which aligns with the agentic aid concept discussed in Section 2. The reviewed approaches primarily rely on text based methods, including prompt engineering and fine tuning, and do not incorporate audio or visual cues. In addition, the study does

[3]The PRISMA methodology (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) provides evidence-based guidelines for transparently reporting systematic reviews by defining a structured process for study identification, screening, and inclusion. [PMB+21]

not address the use of retrieval augmented generation, which based on the arguments presented in Section 2 could provide stronger contextual grounding in domain specific knowledge and enable a more modular and extensible system architecture. Nevertheless, drawing on the literature review and the proposed 3M Framework, it can be inferred that integrating multimodal audio and visual language models together with retrieval augmented generation could further enhance operator support, even though full operator replacement is not considered.

From Promising Capability to Pervasive Bias: Assessing Large Language Models for Emergency Department Triage [LSB+25]

Lee et al. investigate the applicability of large language models (LLMs) for emergency department (ED) triage as high-level cognitive collaborators for professionals. The study is motivated by the limitations of both human triage, which is subject to cognitive bias and variability under pressure, and traditional machine learning approaches, which rely heavily on structured inputs and often fail under distribution shifts.

The authors position LLMs as decision support tools rather than autonomous decision-makers, capable of processing unstructured clinical text and generating triage predictions and rationales.

Method

The study employs a two-part experimental design. First, the authors benchmark LLMs for emergency department triage, evaluating robustness to missing data and distribution shifts across three datasets: MIMIC-IV-ED hospital records, KTAS-labeled hospital data, and curated ESI handbook cases. These datasets combine structured clinical variables with free-text chief complaints. LLM-based approaches, including zero-shot, chain-of-thought, few-shot prompting, retrieval-based in-context learning (KATE), and supervised fine-tuning, are compared against traditional machine learning baselines and a BioBERT-based model using accuracy, F1-score, quadratic-weighted kappa, and mean squared error. Second, a counterfactual analysis systematically perturbs patient sex and race to assess demographic biases in LLM triage predictions using statistical significance testing.

Result

The results show that LLM-based approaches outperform traditional machine learning baselines across most datasets, particularly when retrieval-based in-context learning is used. The KATE method achieves the strongest performance, emphasizing the importance of retrieving relevant examples, while fine-tuned smaller models remain competitive with sufficient domain data. However, notable limitations are identified. LLMs show calibration issues by hesitating to assign the highest acuity level, and the counterfactual analysis reveals significant demographic biases across intersections of sex and race. Although chain-of-thought prompting reduces some bias, it does not eliminate it.

Analysis

In their evaluation, the authors compare LLM-based approaches against traditional machine learning methods and established rule-based triage schemes, highlighting the advantages of LLMs in handling unstructured clinical text. An interesting extension of this work would be an analysis that combines agentic system designs with classical machine learning approaches, as recent studies in other domains suggest that such hybrid architectures can achieve improved accuracy and robustness.

3.3 Industrial Operations

Industrial operations represent an early proving ground for LLM-enabled human–robot collaboration workflows, where operator intent can be translated into contextual tool use, perception, and execution. This section presents a paper that explores the potential application of LLM-based technologies in the context of human–machine interaction (HMI).

Human–robot collaborative visual inspection with Large Language Models [Tas26]

Tasneem et al. present a human–robot collaborative (HRC) visual inspection system that integrates large language models (LLMs) to enable intuitive, speech-based interaction between human operators and industrial robots. The work is motivated by the limitations of traditional GUI- and teach-pendant-based robot programming, which can be cognitively demanding, inflexible, and costly in environments that require frequent reconfiguration. The system uses LLMs to translate natural language instructions into robot inspection actions via deterministic local control. It supports human inspectors by reducing cognitive load and aiding decision-making in complex Industry 5.0 inspection tasks.

4 Conclusion

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